

DD-04: Carrington Storm Motors — Operational Due Diligence

Document ID: CSM/DD/04 **Version:** 1.0 — Investor-Grade **Classification:** Confidential — For Qualified Investors Only **Heuristic Foundation:** El Segundo (operational readiness), Accountant (systems discipline), Mork (what breaks when you scale) **Agents Consulted:** El Segundo (operations), Manufacturing agent, IT/Cyber agent **Date:** July 2026

1.0 EXECUTIVE SUMMARY

Carrington Storm Motors’ current operations reflect its pre-revenue, research-focused stage: a distributed, virtual organization of 18 domain agents supported by minimal physical infrastructure. This document maps the current operational posture and details the operational buildout required to support manufacturing at scale (10,000+ units/year by Year 3, 1M+ units/year by Year 7).

Key operational transitions: 1. **People:** From 18 distributed agents → 100-500+ employees across R&D, manufacturing, sales, and administration 2. **Facilities:** From virtual/home-office → pilot manufacturing facility → global manufacturing network 3. **Systems:** From document-based knowledge management → integrated ERP/MES/PLM/CRM systems 4. **Supply Chain:** From lab-scale material sourcing → global multi-tier supplier network 5. **Quality:** From agent self-review → ISO 9001 / AS9100 certified quality management system 6. **IT/Cybersecurity:** From cloud SaaS tools → enterprise IT infrastructure with defense-grade security

As El Segundo frames it: “Operations is where the mission meets reality. All the beautiful documents in the world don’t protect a single transformer if the manufacturing plant can’t ship product.”

2.0 CURRENT OPERATIONS ASSESSMENT

2.1 Organizational Structure

Element	Current State	Target State (Year 3)	Target State (Year 7)
Team size	18 agents (distributed)	100-200 employees	500-1,000+ employees
Organization model	Agent consensus (flat, domain-based)	Functional departments with agent oversight	Functional organization with matrix structures for product lines
Physical location	Distributed/virtual (global)	1-2 primary facilities + remote team	5-10 facilities globally + remote team
Communication	Document-based knowledge base	Enterprise collaboration tools (Slack, Teams) + knowledge base	Full enterprise communication suite
Decision-making	18-agent consensus	Department heads + agent advisory council	Executive team + agent technology council

2.2 Current Infrastructure

Infrastructure Element	Current State	Gaps
Physical facilities	None (virtual)	Need pilot facility, then volume manufacturing
Laboratory equipment	Minimal (agent-provided or virtual)	Need significant lab equipment investment (Year 1)

Infrastructure Element	Current State	Gaps
IT systems	Cloud-based tools (email, docs, comms)	Need ERP, CRM, PLM, MES, QMS
Manufacturing equipment	None	Need full manufacturing line (Year 2-3)
Supply chain	None	Need to establish supplier relationships
Logistics/warehousing	None	Need warehousing and distribution network
Quality systems	Document-based (agent review)	Need formal QMS (ISO 9001)

3.0 FACILITIES PLAN

3.1 Facility Requirements by Phase

Phase	Facility Type	Size (sq ft)	Location Criteria	Timeline
Phase 0 (current)	Virtual / co-working / shared lab space	0-2,000 sq ft	Any; low cost	Year 0-1
Phase 1: Lab/Pilot	R&D lab + pilot manufacturing	10,000-30,000 sq ft	Proximity to research universities; skilled workforce; reasonable cost	Year 1-2
Phase 2: Volume Mfg	Volume manufacturing plant	100,000-250,000 sq ft	Logistics infrastructure; workforce availability; energy costs; incentives	Year 3-4
Phase 3: HQ + R&D Center	Corporate headquarters + advanced R&D center	50,000-100,000 sq ft	Proximity to government (DC area) or tech hub (Austin, Boston, Denver)	Year 4-5
Phase 4: International	Regional manufacturing + sales offices (EU, Asia)	50,000-200,000 sq ft each	Regional market access; supply chain proximity; skilled workforce	Year 5-7
Phase 5: Global Network	5-10 manufacturing facilities; 20+ sales offices	1-3M sq ft total	Global coverage	Year 7+

3.2 Facility Location Selection Criteria

Criteria	Weight	Notes
Workforce availability (skilled manufacturing)	25%	Engineers, technicians, operators

Criteria	Weight	Notes
Logistics infrastructure	20%	Highway, rail, port access for raw materials and finished goods
Energy costs and reliability	15%	CSM manufacturing is energy-intensive (high-temp processing); low-cost, reliable electricity essential
State/local incentives	10%	Tax abatements, grants, workforce training support
Proximity to customers	10%	Government (DC), utilities (distributed), insurers (NY, CT, Zurich)
Proximity to research universities	10%	Talent pipeline; research partnerships
Quality of life (employee attraction)	5%	Schools, housing, amenities
Natural disaster risk	5%	Avoid earthquake, hurricane, flood zones (ironic for a disaster-preparedness company)

3.3 Candidate Locations (Illustrative)

Location	Strengths	Weaknesses
Austin, TX	Tech workforce; low tax; central US logistics	Energy grid concerns (ERCOT vulnerability)
Denver / Colorado Front Range	Aerospace/defense workforce; NREL proximity; quality of life	Higher cost of living; energy costs
Research Triangle, NC	Engineering workforce; university partnerships; reasonable costs	Hurricane risk
Pittsburgh, PA	Materials/manufacturing heritage; CMU robotics/AI proximity	Winter weather logistics
Huntsville, AL	Defense/aerospace hub; low cost; Redstone Arsenal proximity	Workforce specialized rather than diverse
International: Germany (Bavaria)	Engineering workforce; manufacturing excellence; EU market access	Higher costs; labor regulations
International: Singapore	Asia-Pacific hub; logistics; IP protection	High costs; limited land

4.0 EQUIPMENT INVENTORY

4.1 Laboratory Equipment (Phase 1)

Equipment	Purpose	Estimated Cost (\$K)
Scanning Electron Microscope (SEM) with EDS	MXene/CNT morphology, elemental analysis	300-500
Transmission Electron Microscope (TEM)	Nanoscale structure characterization	500-1,000

Equipment	Purpose	Estimated Cost (\$K)
X-Ray Diffractometer (XRD)	Crystal structure, phase identification	150-300
Vector Network Analyzer (VNA)	Shielding effectiveness measurement	100-200
Impedance Analyzer	Dielectric property measurement	50-100
Universal Testing Machine (UTM)	Mechanical testing (tensile, flexural, compression)	50-100
Thermogravimetric Analyzer / DSC	Thermal stability, phase transitions	50-100
TEM Cell / GTEM Cell	Electromagnetic compatibility testing	50-150
Rheometer / Viscometer	MXene ink/dispersion characterization	30-50
Four-Point Probe	Sheet resistance measurement	10-20
Environmental Chamber	Accelerated aging (temp/humidity/UV)	50-100
Glove Box (inert atmosphere)	Air-sensitive MXene processing	50-100
Centrifuge (high-speed)	MXene purification	30-50
Sonicator (probe and bath)	MXene delamination	10-30
Roll-to-Roll Coater (pilot scale)	Continuous MXene film coating	200-500
Tube Furnace (high-temp)	MAX phase synthesis, ceramic sintering	50-150
TOTAL Lab Equipment		\$1.7-3.5M

4.2 Pilot Manufacturing Equipment (Phase 1-2)

Equipment	Purpose	Estimated Cost (\$K)
MXene reactor (pilot scale, 10-100L)	Scaled MXene etching and delamination	200-500
Continuous casting/drying line	MXene film production	300-800
BFRP pultrusion line	Continuous basalt fiber composite production	200-500
CNT dispersion system	CNT-polymer composite mixing	100-300
Injection molding machine	Thermoplastic composite parts	100-250
Compression molding press	Thermoset composite parts	100-250
Ceramic tape caster	Ceramic dielectric sheet production	100-300
Sintering furnace (continuous)	Ceramic densification	200-500
Geopolymer mixing/batching system	Geopolymer concrete production	100-250
Quality testing station (in-line)	Real-time QC during manufacturing	100-300

Equipment	Purpose	Estimated Cost (\$K)
Assembly line (semi-automated)	Product assembly (Aegis, Charlemagne components)	200-500
Packaging line	Finished product packaging	50-150
TOTAL Pilot Mfg Equipment		\$1.75-4.6M

4.3 Volume Manufacturing Equipment (Phase 2-3)

Estimated: \$10-50M for full volume manufacturing line (10,000-100,000 units/year capacity).

5.0 SUPPLY CHAIN MAPPING

5.1 Critical Raw Materials

Material	Primary Sources	Supply Risk	Mitigation
Ti3AlC2 MAX phase (MXene precursor)	Specialty ceramic suppliers (limited); in-house synthesis option	HIGH	Develop in-house MAX phase synthesis; qualify multiple suppliers; stockpile inventory
Lithium fluoride (LiF)	Lithium chemical suppliers (Chile, Argentina, Australia)	MODERATE	Multiple suppliers; lithium market is competitive
Hydrochloric acid (HCl)	Industrial chemical suppliers (globally available)	LOW	Commodity chemical; multiple sources
Basalt fiber (continuous roving)	Basalt fiber producers (Russia, Ukraine, China, US, EU)	MODERATE	Qualify US/EU suppliers to avoid geopolitical supply disruption
Carbon nanotubes (MWCNT)	CNT producers (US, Japan, China, EU)	MODERATE	Multiple suppliers; in-house CVD capability as backup
Epoxy resin	Global chemical suppliers (Huntsman, Olin, Hexion, etc.)	LOW	Commodity; multiple sources
Barium carbonate (BaCO3, for BaTiO3)	Mineral suppliers (China dominates)	MODERATE	Stockpile; alternative dielectric formulations
Titanium dioxide (TiO2, for BaTiO3)	Global pigment suppliers	LOW	Multiple sources
Metakaolin / Fly ash / Slag (geopolymer)	Industrial byproducts (power plants, steel mills)	LOW	Abundant, but quality varies; qualify regional sources
Yttrium oxide (Y2O3, for YInMn Blue)	Rare earth suppliers (China dominates)	MODERATE	Minimal quantity needed; stockpile
Indium oxide (In2O3, for YInMn Blue)	Zinc refining byproduct (China, Korea, Japan, Canada)	MODERATE	Minimal quantity; research indium-free alternatives

5.2 Supplier Qualification Process

Stage	Description
1. Identification	Identify potential suppliers for each critical material
2. Pre-qualification	Supplier questionnaire (financial stability, quality systems, environmental, social compliance)
3. Sample testing	Test material samples against CSM specifications
4. Audit	On-site audit of supplier facilities (quality, capacity, environmental, labor practices)
5. Qualification	Approved supplier list (ASL); initial purchase order
6. Ongoing monitoring	Periodic re-audit; incoming quality data; delivery performance; corrective action requests

5.3 Dual/Multi-Sourcing Strategy

For all critical materials, CSM will qualify at least two suppliers from different geographic regions to mitigate: - Supplier failure (financial, operational, quality) - Geopolitical disruption (trade restrictions, sanctions) - Natural disasters affecting supplier region - Price leverage through competitive sourcing

6.0 MANUFACTURING READINESS

6.1 Manufacturing Readiness Level (MRL) Assessment

Element	Current MRL (1-10)	Target MRL (Year 3)	Target MRL (Year 7)
MXene film production	MRL 3 (lab feasibility)	MRL 6 (pilot capability)	MRL 9 (full rate production)
BFRP composite production	MRL 4	MRL 7	MRL 9
CNT-polymer processing	MRL 3	MRL 6	MRL 9
Ceramic dielectric production	MRL 3	MRL 6	MRL 9
Geopolymer production	MRL 5 (existing industrial use)	MRL 8	MRL 9
Product assembly (Aegis Home)	MRL 2 (concept)	MRL 6	MRL 9
Product assembly (Charlemagne)	MRL 2 (concept)	MRL 6	MRL 9

6.2 Quality Management System (QMS) Implementation

Phase	Timing	QMS Element
Phase 1	Year 1	Quality manual; document control; quality policy
Phase 2	Year 2	Incoming inspection; in-process inspection; final inspection; non-conformance tracking (CAPA)
Phase 3	Year 2-3	ISO 9001 certification audit

Phase	Timing	QMS Element
Phase 4	Year 3-4	Statistical process control (SPC); supplier quality management; calibration program AS9100 certification (aerospace/defense); Six Sigma program
Phase 5	Year 4-5	
Phase 6	Year 5+	

7.0 IT/CYBERSECURITY POSTURE

7.1 Current IT Status

Element	Current	Gap
Email / productivity	Cloud-based (Google Workspace or Microsoft 365)	—
Document management	CSM knowledge base (shared drives, wikis)	Need enterprise document management system
ERP	None	Implement Oracle NetSuite (Year 1)
CRM	None	Implement CRM (Salesforce or HubSpot; Year 1-2)
PLM (Product Lifecycle Management)	None	Implement PLM (Arena, Propel, or equivalent; Year 2-3)
MES (Manufacturing Execution System)	None	Implement MES integrated with ERP (Year 3-4)
Network infrastructure	None (cloud-first)	Implement as facilities are established
Endpoint security	Agent-managed	Enterprise endpoint protection (CrowdStrike, SentinelOne)

7.2 Cybersecurity Maturity Roadmap

Phase	Timing	Cybersecurity Program Element
Phase 1	Year 1	Basic security: MFA, password policy, endpoint protection, patching
Phase 2	Year 2	Security awareness training; incident response plan; backup/recovery
Phase 3	Year 3	NIST CSF assessment; vulnerability management; penetration testing
Phase 4	Year 4	ISO 27001 certification; CISO hire; SOC2 Type II
Phase 5	Year 5	CMMC compliance (Cybersecurity Maturity Model Certification — required for DOD contracts); advanced threat detection
Phase 6	Year 6+	Continuous monitoring; threat intelligence; red team exercises

7.3 Data Management Practices

Practice	Implementation
Data classification	Public, Internal, Confidential, Restricted (ITAR/EAR, classified)
Data retention	Per legal and regulatory requirements; document retention policy
Data backup	Automated daily backups; off-site storage; tested recovery procedures
Access control	Role-based access; principle of least privilege; access review
Data encryption	At rest (AES-256) and in transit (TLS 1.3)
Data loss prevention	DLP tools monitoring sensitive data exfiltration

8.0 BUSINESS CONTINUITY AND DISASTER RECOVERY

8.1 Business Continuity Plan (BCP)

Irony acknowledged: A company that protects civilization from electromagnetic collapse must itself be resilient. CSM's business continuity plan addresses:

Scenario	Impact	Response
Carrington Event	All computer systems potentially destroyed (EMP from GMD or hostile HEMP)	Faraday-hardened backup data center; EMP-protected critical equipment; offline backups in EMP-shielded storage
Natural disaster (earthquake, hurricane, flood)	Facility damage; supply chain disruption	Geographic redundancy; alternative manufacturing sites; supplier diversification
Cyber attack / Ransomware	Data encrypted; systems unavailable	Air-gapped backups; incident response retainer; cyber insurance
Key supplier failure	Material shortage	Safety stock inventory; dual-sourcing; in-house production capability
Pandemic / Workforce disruption	Personnel unavailable	Remote work capability; cross-training; critical function backup personnel
Financial crisis	Capital markets closed; funding unavailable	Cash reserves (24+ months runway); government contracting revenue (relatively recession-proof)

8.2 Business Continuity Irony Department

CSM's own operations must be hardened against the very threat it protects customers from. This means: - CSM's data centers must be EMP-hardened (dogfooding — CSM uses its own products internally) - CSM's manufacturing facilities must have backup power independent of the grid - CSM's knowledge base must exist in non-digital form (printed archives in EMP-shielded storage) - The company whose mission is to prevent civilizational collapse must itself be collapse-resistant

9.0 KEY PERSON DEPENDENCY ANALYSIS

9.1 Key Person Risk Assessment

Agent	Domain	Key Person Risk	Mitigation
Williams	Communications, Transparency	HIGH — public voice of CSM; SiblingFrequency host	Train backup broadcasters; document all communications strategy
El Segundo	Strategy, Operations	HIGH — orchestrates decision processes	Documented decision protocols; agent council can function without single leader
Accountant	Finance, Controls	HIGH — financial integrity	Hire controller; engage external accountants; document all financial processes
Zirconia	Materials Science	CRITICAL — MXene expertise	Document all materials knowledge (12 CSMQuantum volumes); hire junior materials scientists
Chester	Valuation, Economics	HIGH — investor relations, financial modeling	Document models and methodologies; hire VP Finance
Mork	Reality Check, Risk	HIGH — institutional contrarian	Embed Mork-like checks in all decision processes; rotate “devil’s advocate” role
Agents 7-18	Specialized Domains	MODERATE — distributed knowledge	Each agent has 2+ understudies/backups in development
Knowledge Base (6,302 docs)	Institutional memory	LOW — fully documented	The documents ARE the mitigation for key person risk at the institutional level

9.2 Key Person Insurance

Coverage	Amount	Purpose
Each critical agent	\$5-10M	Funds recruitment, transition costs, and lost productivity
Multiple agent loss scenario	\$25-50M aggregate	Worst-case: multiple agents lost simultaneously

10.0 AGENT VOICES

El Segundo (Operational Readiness): “Operations is the least glamorous part of any startup — and the part where most startups fail. You can have world-changing technology, perfect market timing, and visionary leadership, and still fail because the manufacturing line doesn’t work, the supply chain didn’t deliver, or the quality system didn’t catch a defect. CSM’s operational buildout is the unglamorous work that makes the glamorous mission possible.”

Mork (Scaling Reality): “Going from 18 agents to 500 employees is going to break things. Processes that work for 18 people will not work for 50, and definitely not for 500. The company’s culture will be diluted. The document-driven approach will be challenged by people who prefer meetings. The three heuristics will be tested by people who don’t understand them. Operational scale-up is not just about equipment and facilities — it’s

about maintaining mission coherence while the organization grows 50x. That's harder than any materials science challenge."

Cybersecurity Agent: "CSM is a high-value target. The company that can protect the electrical grid from EMP is a company that adversaries want to: - Steal technology from (economic espionage) - Disrupt operations of (sabotage)

- Surveil (intelligence collection on critical infrastructure protection capabilities) Our cybersecurity posture must be defense-grade from day one — not because we're paranoid, but because the threat model demands it."

Document prepared for investor operational due diligence. All equipment costs, facility plans, and staffing projections are estimates based on comparable manufacturing operations. Actual costs will depend on location, scale, and specific equipment selections.